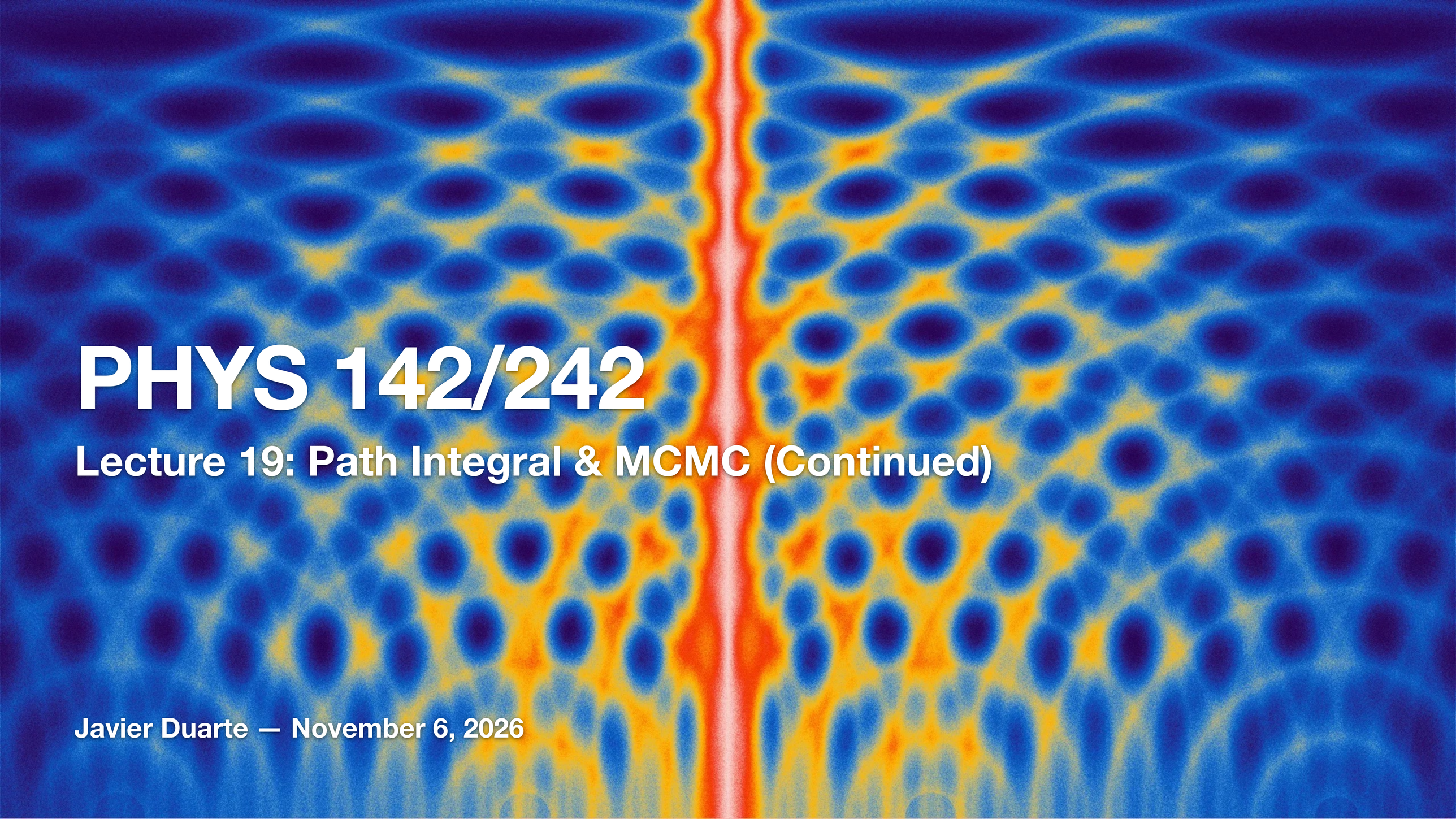


PHYS 142/242

Lecture 19: Path Integral & MCMC (Continued)

Javier Duarte — November 6, 2026

Quiz 4: Monday 11/9, in class (30 min)

- No devices, but basic calculator is allowed; your lowest quiz score is dropped
- **Covers Lectures 14-15**: Markov chains and the Metropolis-Hastings algorithm
  - Transition probabilities and the transition matrix (each column sums to 1)
  - Stationary distributions: solving for a small chain
  - Detailed balance
  - Metropolis vs. Metropolis-Hastings acceptance; symmetric vs. asymmetric proposals
- **Formula sheet provided**: the definitions of the transition matrix, stationarity and detailed balance, and the Metropolis and Metropolis-Hastings acceptance probabilities.
- **Practice**: for a small chain, build $P(m \mid n) = q(m \mid n)A(m \mid n)$ from a given proposal $q(m \mid n)$ and acceptance $A(m \mid n)$, then solve for its stationary distribution $P\pi = \pi$. Use the convention that $P(m \mid n)$ sits in row m, column n
- **Not on Quiz 4**: Lectures 16-19 (autocorrelation and error bars, Ising, path-integral MCMC). Those are covered by Assignment 3 and Oral Checkpoint 3.

Quiz 4 Formula Sheet

$P(m\,|\,n)$ = prob. of moving from n to m, $\qquad \sum_m P(m\,|\,n) = 1, \qquad p^{(k+1)} = P\,p^{(k)}$

stationary: $P\pi = \pi \;\Leftrightarrow\; \sum_n P(m\,|\,n)\,\pi_n = \pi_m$ (unique for irreducible, aperiodic chain)

detailed balance: $P(m\,|\,n)\,\pi_n = P(n\,|\,m)\,\pi_m \;\Rightarrow\;$ π is stationary

$$P(m\,|\,n) = q(m\,|\,n)\,A(m\,|\,n)\ (m \neq n), \qquad P(n\,|\,n) = 1 - \sum_{m \neq n} P(m\,|\,n)$$

$$A_{\text{Metropolis}} = \min\left(1, \frac{\pi_m}{\pi_n}\right), \qquad A_{\text{Hastings}} = \min\left(1, \frac{\pi_m\,q(n\,|\,m)}{\pi_n\,q(m\,|\,n)}\right)$$

Recap: Path integral for quantum stat. mech.

Setting $x_b = x_a = x$ and integrating over x, with $\tau_a = 0$, and $\tau_b = \hbar\beta$ where $\beta = 1/(k_{\mathrm{B}}T)$, gives the partition function

$$Z = \int \mathscr{D}x(\tau) \exp\left[-S_E/\hbar\right] \qquad S_E = \oint_0^{\hbar\beta} L_E\big(x(\tau)\big)\, d\tau \qquad L_E(x(\tau)) = \frac{m}{2}\left(\frac{dx}{d\tau}\right)^2 + V(x(\tau))$$

Discretize "time" with N increments. Each vector $(x_0, x_1, \ldots, x_{N-1})$, where $x_N \equiv x_0$, in N-dimensional space represents an imaginary-time path.

$$S_E \approx \sum_{i=0}^{N-1} \left[\frac{m\,(x_{i+1} - x_i)^2}{2\eta} + \eta\, V(x_i)\right] \qquad p(x_0, \ldots, x_{N-1}) = \frac{1}{Z}\left(\frac{2\pi\hbar\eta}{m}\right)^{-N/2} e^{-S_E/\hbar}$$

Recap: MCMC steps

Begin with an initial path, which may be an array of random numbers (‘hot’ start) or zeros (‘cold’ start).

Update the path by applying Metropolis algorithm to each element x_i of the path in random order (called a ‘sweep’)

(a) Generate a uniform random number $u \in [-h, h]$ (where h is the ‘hit size’)

(b) Propose the new value $x_i' = x_i + u$ of the path element and calculate the resulting change in the action: $\Delta S_E = \frac{m}{\eta}(x_i' - x_i)(x_i' + x_i - x_{i-1} - x_{i+1}) + \eta\,[V(x_i') - V(x_i)]$

(indices wrap around the ring $x_{-1} \equiv x_{N-1}; x_N \equiv x_0$)

If $\Delta S_E \leq 0$, accept the new path element.

If $\Delta S_E > 0$, accept with probability $\exp(-\Delta S_E/\hbar)$

Recap: finite β

At any β, the chain samples the thermal distribution

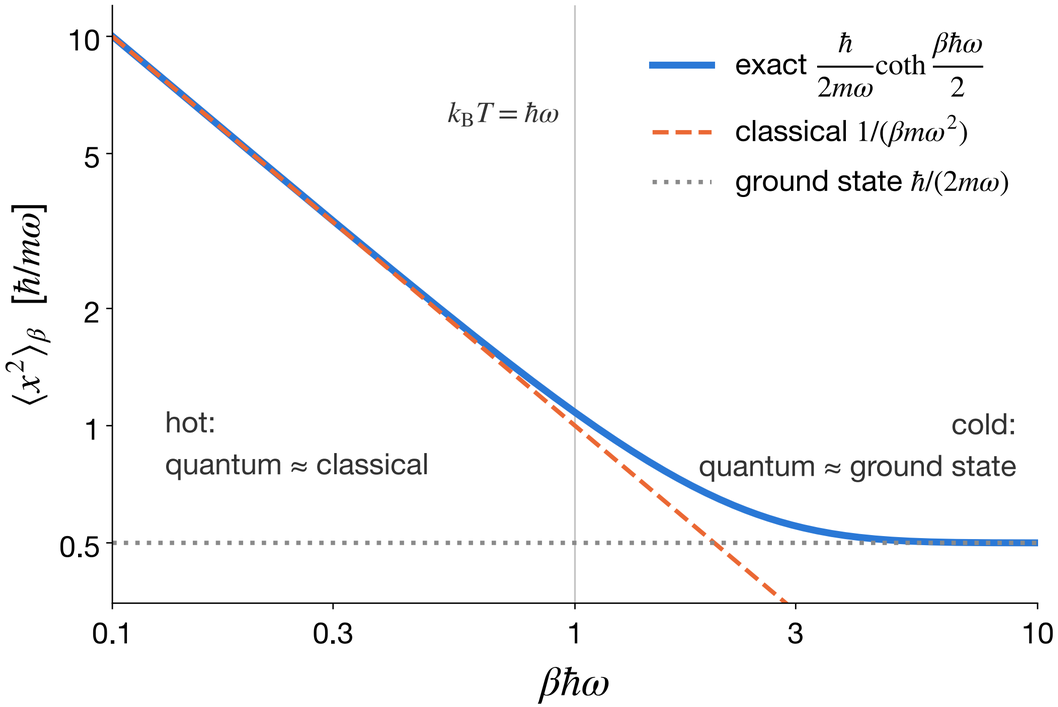
$$\frac{\rho(x,x)}{Z} = \sum_n p_n \,|\,\phi_n(x)\,|^2, \qquad p_n = \frac{e^{-\beta E_n}}{Z}$$

For harmonic oscillator, estimator is

$$\langle x^2 \rangle_\beta \approx \left\langle \frac{1}{N} \sum_{i=0}^{N-1} x_i^2 \right\rangle_{\text{sweeps}}$$

Two relevant limits:

$$\langle x^2 \rangle_\beta = \frac{\hbar}{2m\omega} \coth \frac{\beta\hbar\omega}{2} \;\to\; \begin{cases} \hbar/(2m\omega) & \text{ground state, } \beta\hbar\omega \gg 1 \\ 1/(\beta m\omega^2) & \text{classical equipartition, } \beta\hbar\omega \ll 1 \end{cases}$$



Equilibration and burn in

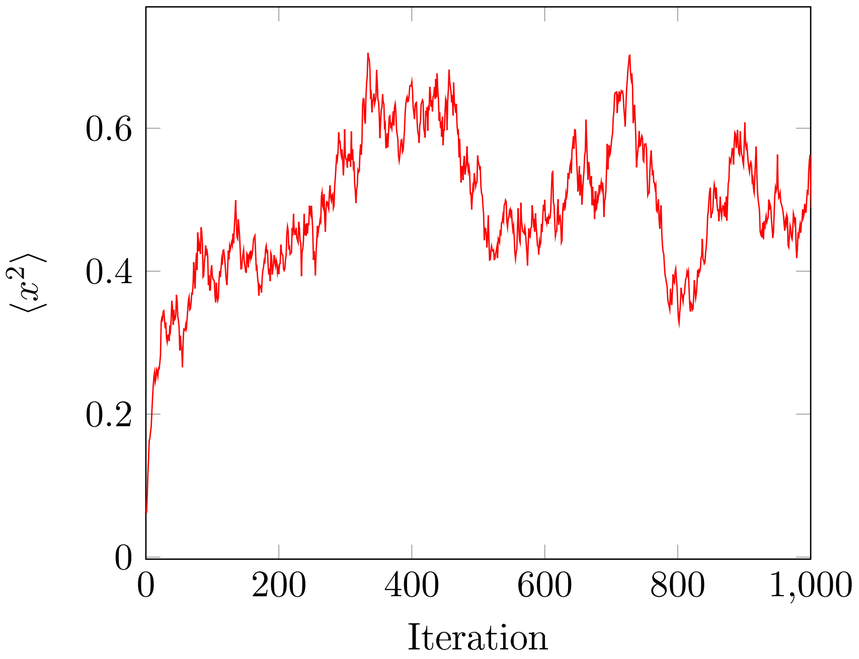


Figure 2.1: The evolution of $\langle x^2 \rangle$ with iterations of the Metropolis algorithm for lattice spacing $a = 0.1$, $\mu^2 = 1$.

R. Rodgers and L. Raes, DESY Summer Student Programme (2014)

Can use $\langle x^2 \rangle$ or equivalently $\langle E \rangle$ (for harmonic oscillator) to measure equilibration and burn in necessary

Key: $a = \eta$, $\Delta = h$, $\mu^2 = \omega^2$

Acceptance rate vs. hit size h

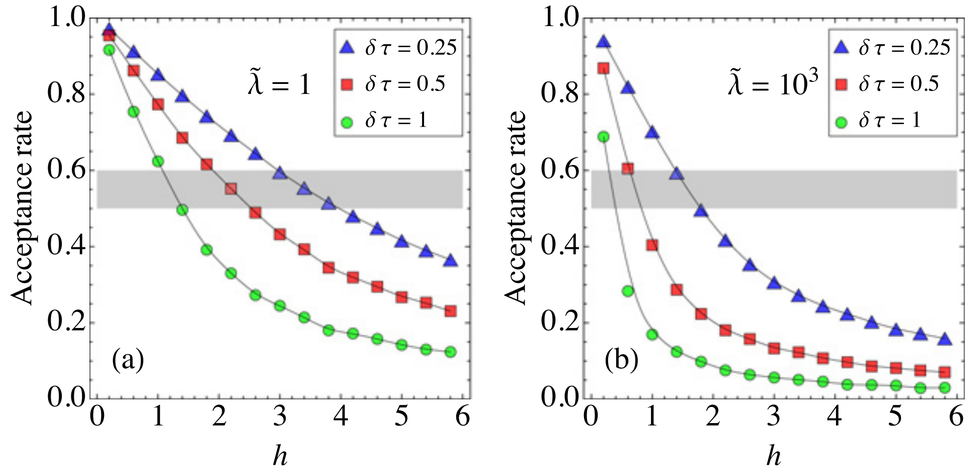


Figure 3. Acceptance rate versus the hit size h for the quantum anharmonic oscillator with quartic coupling constant (a) $\tilde{\lambda} = 1$ and (b) $\tilde{\lambda} = 1000$ for the indicated discretizations. The target of 50%−60% acceptance rate is indicated by shading. The curves are spline fits to the data.

S. Mittal et al., Eur. J. Phys. 41 055401 (2020)

Acceptance rate depends on hit size h and time slice η

We target 40-60% in Assignment 3

Autocorrelation function

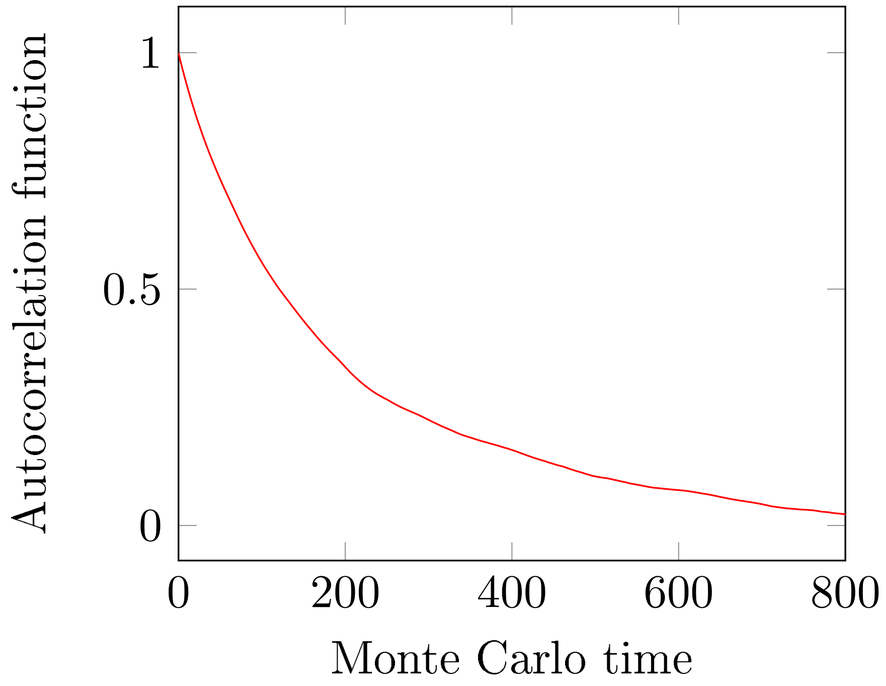


Figure 3.2: The autocorrelation function for $a = 0.05$, $\mu^2 = 1$, $m = 1$.

Key: $a = \eta$, $\Delta = h$, $\mu^2 = \omega^2$

R. Rodgers and L. Raes, DESY Summer Student Programme (2014)

Measure autocorrelation function

$$\rho(t) = \frac{\langle (f_i - \bar{f})\,(f_{i+t} - \bar{f})\rangle}{\langle (f_i - \bar{f})^2\rangle}$$

$$\approx e^{-t/\tau_{\text{exp}}}$$

$$\tau_{\text{int}} = 1 + 2\sum_{t\geq 1}\rho(t) \approx 2\,\tau_{\text{exp}}$$

Finds $\tau_{\text{exp}} \approx 190$, but it depends on η, etc.

Autocorrelation time vs. hit size

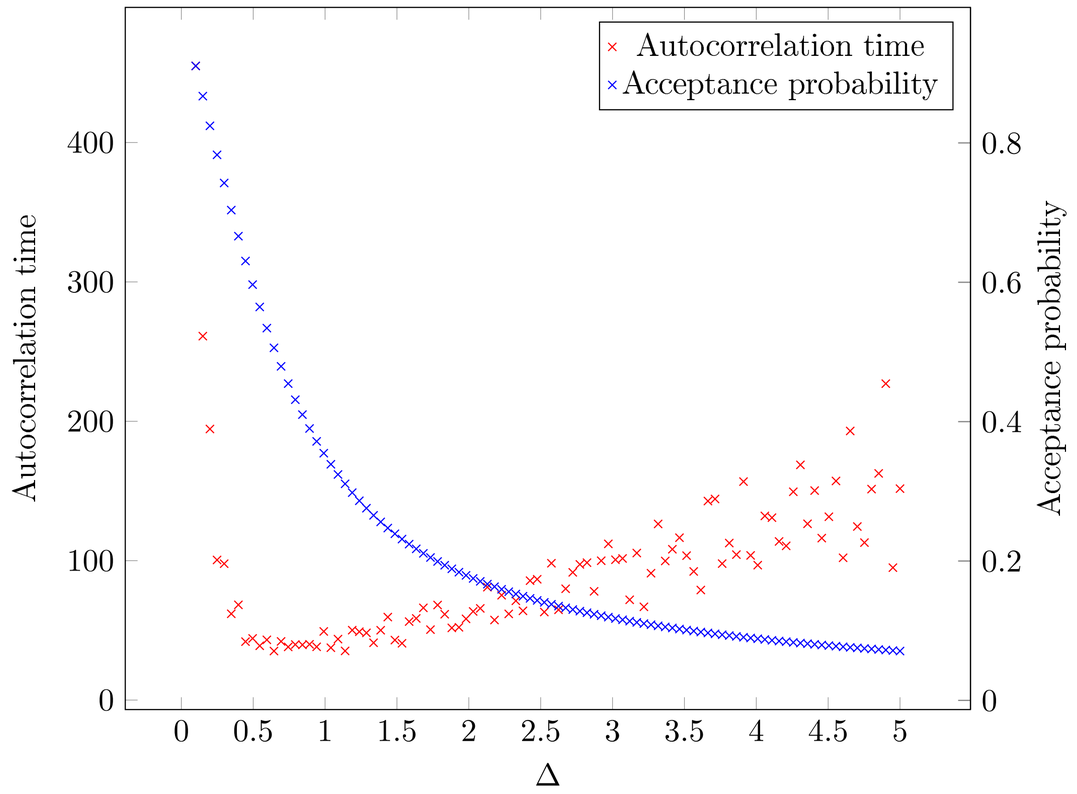


Figure 3.6: The autocorrelation time and the fraction of accepted values of x'_j as a function of Δ for $a = 0.1$, $\mu = 1$, $m = 1$.

Key: $a = \eta$, $\Delta = h$, $\mu^2 = \omega^2$

R. Rodgers and L. Raes, DESY Summer Student Programme (2014)

Autocorrelation time τ_{int} and acceptance rate both depend on hit size h

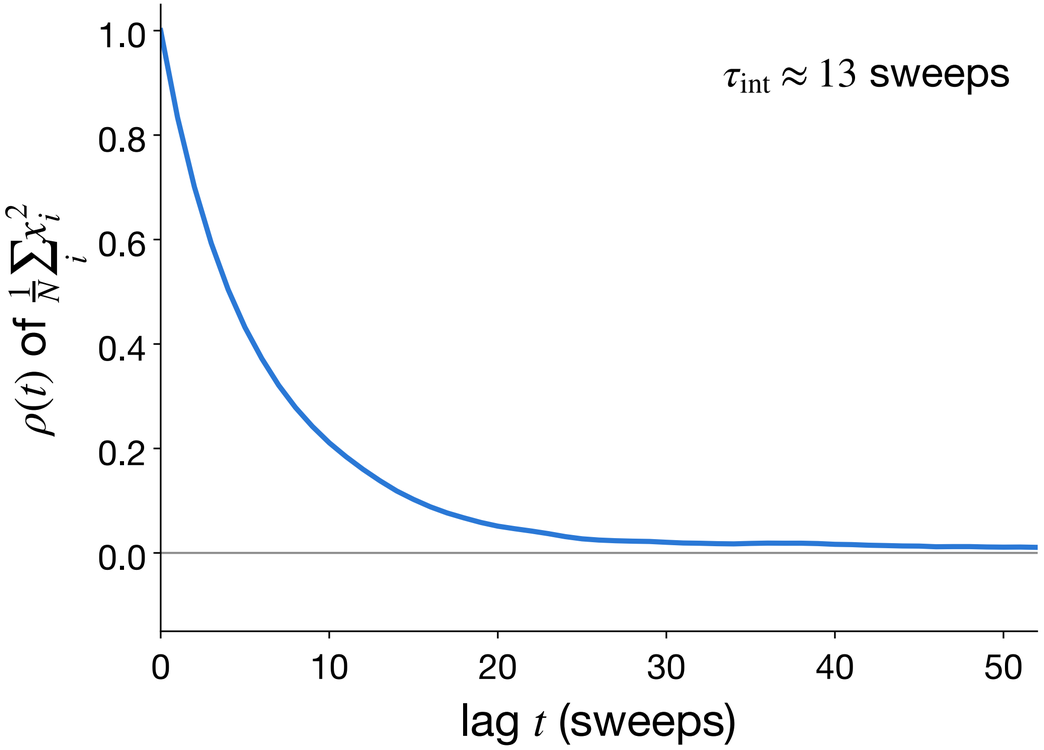
Creutz and Freedman choose $h = 2\sqrt{\eta}$, and for $\eta = 0.1$, this gives $h = 0.63$ and acceptance rate $\sim 50\,\%$

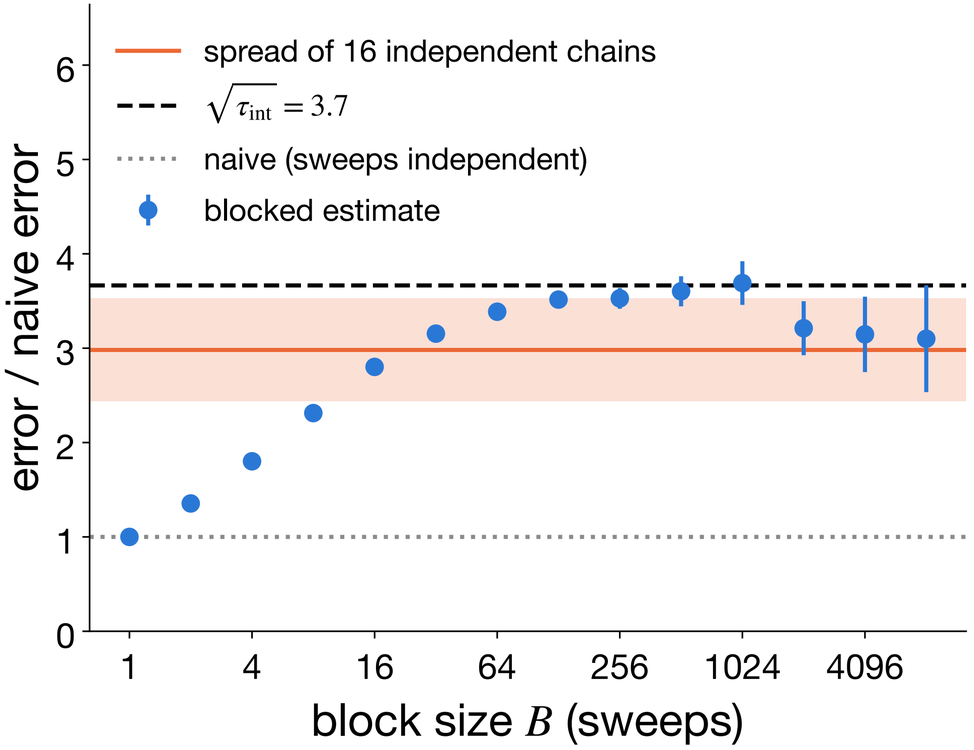
Note: τ_{int} is nearly flat around its minimum, from about 30% to 55% acceptance.

Naive error bars

- For $\frac{1}{N}\sum_i x_i^2$, $\tau_{\text{int}} \approx 13$ sweeps, *so naive error is too small by* $\sqrt{\tau_{\text{int}}} \approx 3.7$.
- 16 independent chains give an error $3.0\times$ the naive one (consistent with 3.7 within the ~20% uncertainty spread)

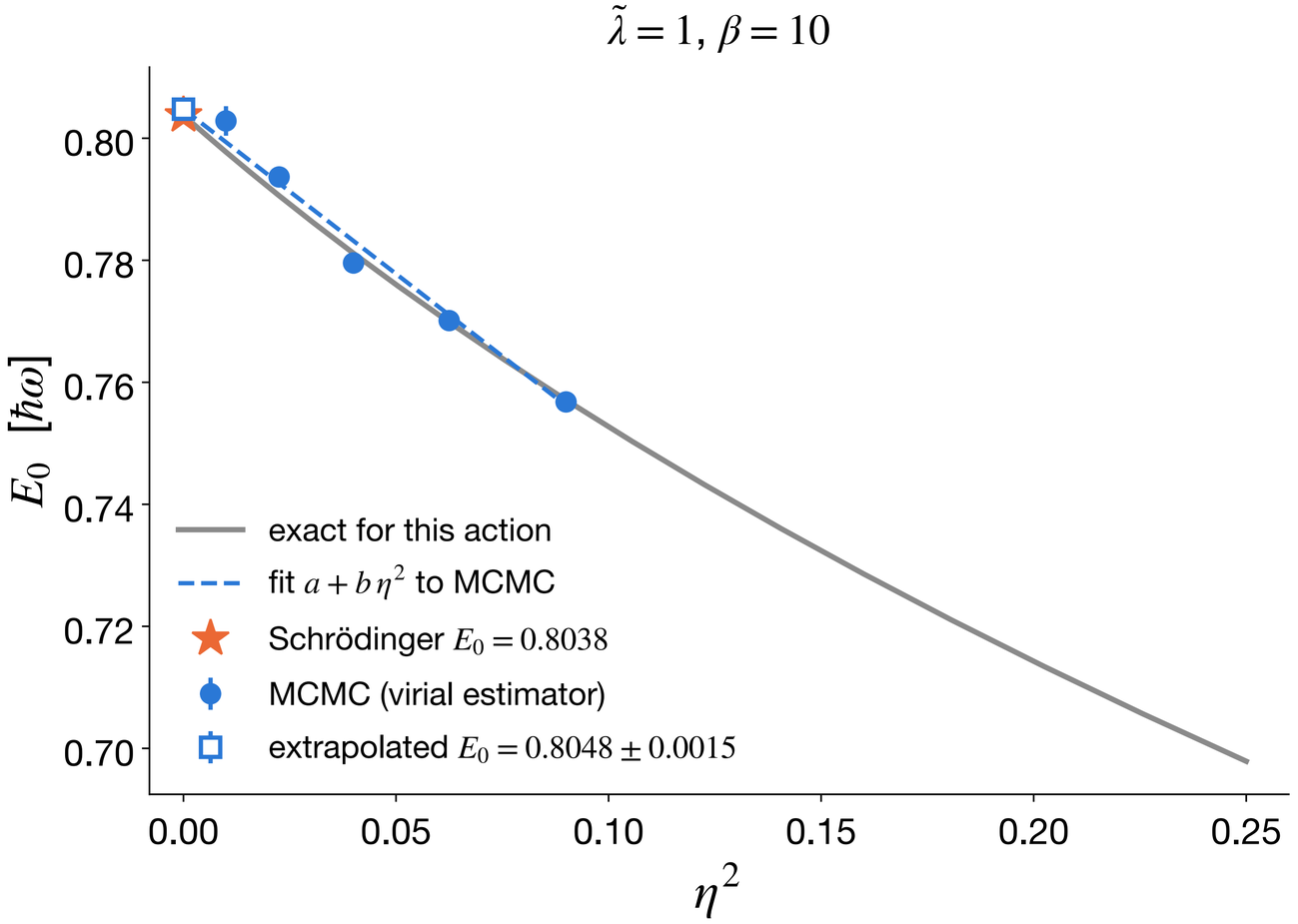
$\tilde{\lambda} = 1,\ \beta = 8,\ \eta = 0.25,\ h = 0.9$ (53% accepted)





Two sources of error

- Stat. error $\propto \sqrt{\tau_{\mathrm{int}}/N_{\mathrm{sweeps}}}$
- Time-step error scales as η^2 as $\eta \to 0$
- Fit of $a + b\eta^2$ to the MCMC points at $\eta \leq 0.3$ gives $E_0 = 0.8048 \pm 0.0015$, versus $E_0 = 0.8038$ from Schrödinger
- Extrapolation only reliable where curve is linear



Imaginary-time correlators

- Consider the imaginary time correlator ($\tau = k\eta$)

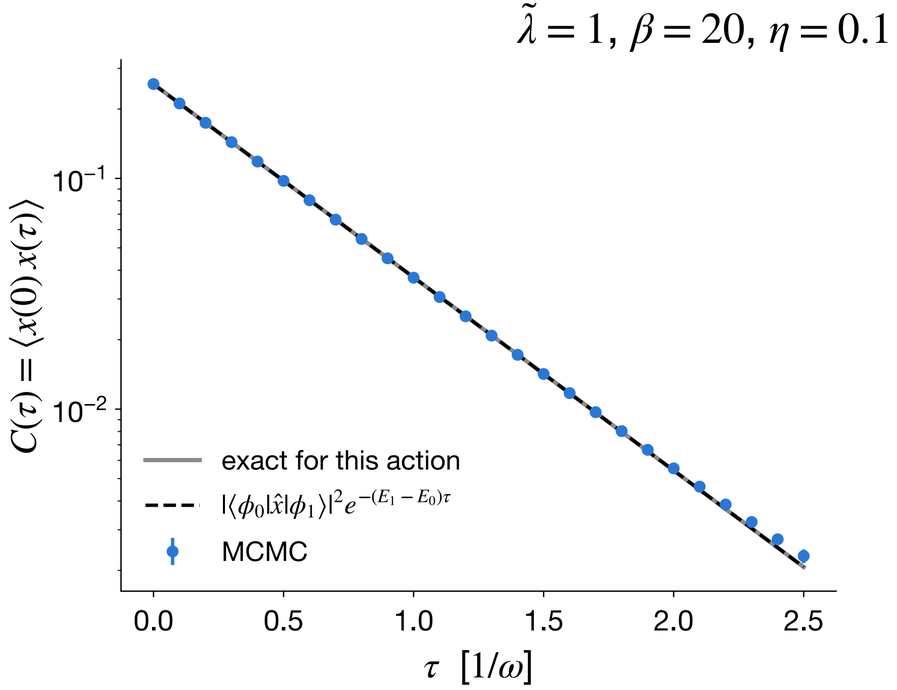
$$C(\tau) = \langle x(0)\, x(\tau) \rangle \approx \left\langle \frac{1}{N} \sum_{i=0}^{N-1} x_i\, x_{i+k} \right\rangle_{\text{sweeps}}$$

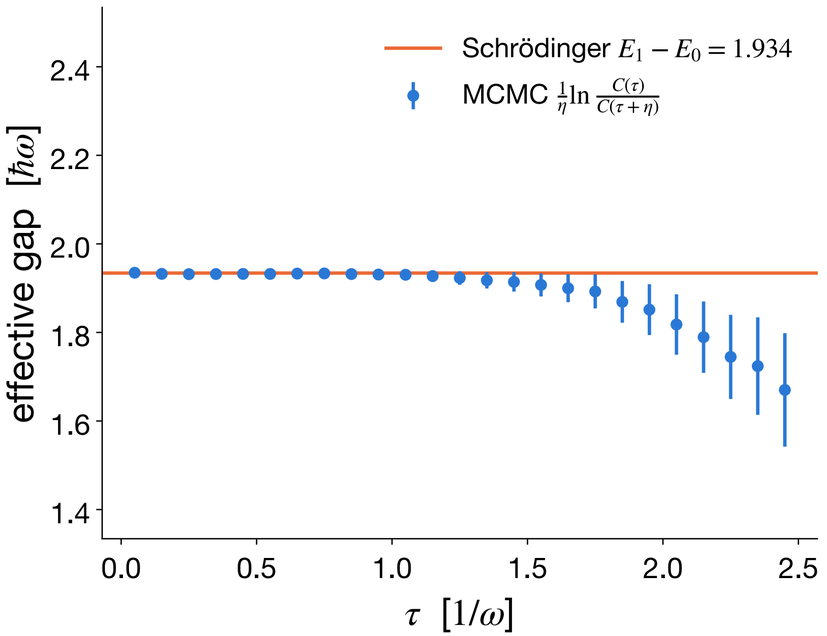
- At low temperature (high β),

$$C(\tau) \xrightarrow{\beta\to\infty} \sum_n |\langle \phi_0 | \hat{x} | \phi_n \rangle|^2 e^{-(E_n - E_0)\tau/\hbar}$$

$$\xrightarrow{\tau \text{ large}} |\langle \phi_0 | \hat{x} | \phi_1 \rangle|^2 e^{-(E_1 - E_0)\tau/\hbar}$$

- Effective gap $-\hbar d \ln C / d\tau$ levels off at $E_1 - E_0 = 1.934\hbar\omega$, while noise grows with τ





Proof

- From the path integral to a trace:

$$C(\tau) = \frac{1}{Z}\int dx_0\, dx_\tau\; x_0\, x_\tau \,\langle x_0|\, e^{-(\hbar\beta-\tau)\hat{H}/\hbar}\,|x_\tau\rangle\langle x_\tau|\, e^{-\tau\hat{H}/\hbar}\,|x_0\rangle = \frac{1}{Z}\,\mathrm{Tr}\left[e^{-(\hbar\beta-\tau)\hat{H}/\hbar}\,\hat{x}\,e^{-\tau\hat{H}/\hbar}\,\hat{x}\right]$$

- Set $\beta \to \infty$ at fixed τ:

$$\frac{e^{-(\hbar\beta-\tau)\hat{H}/\hbar}}{Z} = \frac{\sum_m e^{-\beta E_m}\, e^{\tau E_m/\hbar}\,|\phi_m\rangle\langle\phi_m|}{\sum_m e^{-\beta E_m}} \xrightarrow{\beta\to\infty} e^{\tau E_0/\hbar}\,|\phi_0\rangle\langle\phi_0| \quad \Rightarrow \quad C(\tau) \to e^{\tau E_0/\hbar}\,\langle\phi_0|\,\hat{x}\,e^{-\tau\hat{H}/\hbar}\,\hat{x}\,|\phi_0\rangle$$

- Insert a complete set of states and take the limit for large τ:

$$C(\tau) \to \sum_n |\,\langle\phi_0|\,\hat{x}\,|\phi_n\rangle\,|^2\, e^{-(E_n-E_0)\tau/\hbar} \xrightarrow{\tau\ \text{large}} |\,\langle\phi_0|\,\hat{x}\,|\phi_1\rangle\,|^2\, e^{-(E_1-E_0)\tau/\hbar}$$

- Effective gap:

$$-\hbar\,\frac{d\ln C}{d\tau} = \frac{\sum_n (E_n - E_0)\, w_n(\tau)}{\sum_n w_n(\tau)}, \qquad w_n(\tau) = |\,\langle\phi_0|\,\hat{x}\,|\phi_n\rangle\,|^2\, e^{-(E_n-E_0)\tau/\hbar}$$

Preview: hard sampling in double well

- The barrier means the path center rarely crosses between wells, so τ_{int} becomes huge
- Same issue as the Ising chain flipping between $\pm m$ below T_c
- What moves could help?

